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Section	**B**
Assignment	**Computer programing**
Submitted to	**Eng. Rizwan**

Assignment No 1 - Full C++ Code Solution

```
#include <iostream>
using namespace std;
int main() {
// i. Declare variables
double voltage, current, resistance;
// ii. Display message
cout << "Welcome to Object Oriented Programming in C++!"
<< endl;
// iii. Read age
int age;
cout << "Enter age: ";
cin >> age;
// iv. Temperature check
double temperature;
cout << "Enter temperature: ";
cin >> temperature;
if (temperature > 100)
```

```
cout << "Overheating detected!" << endl;
// v. Area of circle
double r, area;
cout << "Enter radius: ";
cin >> r;
area = 3.14159 * r * r;
cout << "Area = " << area << endl;
// vi. Print x, y, z
int x, y, z;
cout << "Enter x, y, z: ";
cin >> x >> y >> z;
cout << x << "," << y << "," << z << endl;
// vii. Sum of voltage and current
cout << "Enter voltage and current: ";
cin >> voltage >> current;
double total = voltage + current;
cout << "Total = " << total << endl;
// viii. Division by zero check
int denominator;
cout << "Enter denominator: ";
cin >> denominator;
if (denominator == 0)
cout << "Error: Division by zero" << endl;
// ix. Increment count
int count;
cout << "Enter count: ";
cin >> count;
count++;
cout << "New count = " << count << endl;
// x. Read three floating numbers
float a, b, c;
cout << "Enter a, b, c: ";
cin >> a >> b >> c;
// xi. Series resistance
double r1, r2, r3;
cout << "Enter three resistances (series): ";
cin >> r1 >> r2 >> r3;
double series = r1 + r2 + r3;
cout << "Total Resistance (Series) = " << series << endl;
// xii. Parallel resistance
cout << "Enter three resistances (parallel): ";
cin >> r1 >> r2 >> r3;
double parallel = 1 / (1/r1 + 1/r2 + 1/r3); cout <<
"Total Resistance (Parallel) = " << parallel << endl;
// xiii. Voltage division rule
double R1 = 10, R2 = 20, V = 12;
double V_R2 = (V * R2) / (R1 + R2);
cout << "Voltage across R2 = " << V_R2 << endl;
// xiv. Current division rule
double Req = (R1 * R2) / (R1 + R2);
double I_total = V / Req;
```

```
double I_R2 = (I_total * R1) / (R1 + R2);
cout << "Current through R2 = " << I_R2 << endl;
// xv. GPA calculation
int n;
cout << "Enter number of subjects: ";
cin >> n;
float totalPoints = 0, totalCredits = 0;
for (int i = 0; i < n; i++) {
float grade, credit;
cout << "Enter grade point and credit hours: ";
cin >> grade >> credit;
totalPoints += grade * credit;
totalCredits += credit;
}
float GPA = totalPoints / totalCredits;
cout << "Semester GPA = " << GPA << endl;
return 0;
}
```